

INTRODUCTION

Key concepts:

- Conditional probability
- Bayes' Theorem
- Prior and posterior distributions
- Likelihood function
- Simple random sampling without replacement
- Selection probabilities
- Sampling distribution
- Confidence and credible intervals

1. INTRODUCTION

Theorem 1 (Bayes). *Suppose x and y are random variables with joint density $p(x, y)$ and marginal densities $p(x)$ and $p(y)$, respectively. Then*

$$p(x|y) = \frac{p(x) p(y|x)}{p(y)}$$

where $p(x|y) = p(x, y)/p(y)$ and $p(y|x) = p(x, y)/p(x)$ are the conditional density functions.

The proof is trivial, so what's the big deal? No one disputes the result. The controversy is about how it is applied. We consider two toy examples before taking up finite population sampling—a “real application” which illustrates the differences between Bayesian and frequentist inference.

1.1. Example: An Inverse Probability Calculation. For an uncontroversial example, suppose we select a person at random and x is their race (Black or White) and y is how they voted (Democratic or Republican). To keep the arithmetic easy, suppose 10% of the population is black, 90% of blacks voted Democratic and 40% of whites voted Democratic. The probability of selecting a Democrat is

$$\begin{aligned} p(\text{Democrat}) &= p(\text{Black}) p(\text{Democrat}|\text{Black}) + p(\text{White}) p(\text{Democrat}|\text{White}) \\ &= 0.1 \times 0.9 + 0.9 \times 0.4 = 45\%. \end{aligned}$$

Applying Bayes' Theorem, we obtain

$$p(\text{Black}|\text{Democrat}) = \frac{p(\text{Black}) p(\text{Democrat}|\text{Black})}{p(\text{Democrat})} = \frac{0.1 \times 0.9}{0.45} = 20\%$$

Bayes' Theorem is sometimes called “The Law of Inverse Probability” because it gives you a way to convert $p(y|x)$ into $p(x|y)$. For a cute visualization, see <http://www.countbayesie.com/blog/2015/2/18/bayes-theorem-with-lego>.

1.2. Example: The Monty Hall Problem. The preceding example involved “objective” probabilities which could be interpreted as relative frequencies. More controversial is when the probabilities are “subjective,” such as those in the Monty Hall problem.¹ The game show host has hidden a valuable prize behind one of three doors, but he doesn’t tell us which one. We choose one of the doors and then the host opens one of the two remaining doors, revealing one that did not contain the prize. He then asks us whether we would like to stick with our original choice or switch to the remaining closed door. What should we do?

We need to calculate the probability that the prize is behind the door we originally selected and the other closed door. Number the doors 1, 2, and 3, and let x denote the door hiding the prize and y the door opened by the host. For concreteness, suppose we initially selected door 1, so either $y = 2$ or $y = 3$. Before the host opens a door, we would assign $p(x) = 1/3$ for $x = 1, 2, 3$. If you’ve watched the show, you know that the host *always* reveals one of the other doors after the contestant has made their initial choice. If $x = 1$, then Monty has two possible doors which he can open, so let’s assume he picks each with equal probability: $p(y = 2|x = 1) = p(y = 3|x = 1) = 1/2$. However, if $x \neq 1$, then Monty has only a single door that he can open without revealing the prize, so $p(y = 2|x = 2) = p(y = 3|x = 3) = 0$ and $p(y = 3|x = 2) = p(y = 2|x = 3) = 1$. Combining these with Bayes’ Theorem gives

$$p(x = 1|y = 2) = \frac{p(x = 1)p(y = 2|x = 1)}{p(y = 2)} = \frac{(1/3) \times (1/2)}{1/2} = \frac{1}{3},$$

since

$$\begin{aligned} p(y = 2) &= p(x = 1)p(y = 2|x = 1) + p(x = 2)p(y = 2|x = 2) + p(x = 3)p(y = 2|x = 3) \\ &= \frac{1}{3} \times \frac{1}{2} + \frac{1}{3} \times 0 + \frac{1}{3} \times 1 = \frac{1}{2}. \end{aligned}$$

(This is the *Law of Total Probability*, which will reappear shortly.) A similar calculation for $x = 3$ gives

$$p(x = 3|y = 2) = \frac{p(x = 3)p(y = 2|x = 3)}{p(y = 2)} = \frac{(1/3) \times 1}{1/2} = \frac{2}{3}.$$

This strikes some people as counter-intuitive because the host’s actions don’t seem to contain any information that would make us think the third door is more likely to contain the prize than the first door. It’s true that the Monty hasn’t altered our beliefs about door number 1 (*i.e.*, $p(x = 1|y = 2) = 1/3$), since he never opens the first door regardless of whether it hides the prize or not. But we now *know* that door number 2 doesn’t contain the prize, so $p(x = 2|y = 2) = 0$. Since there are only three alternatives and the probabilities must sum to one ($p(x = 1|y = 2) + p(x = 2|y = 2) + p(x = 3|y = 2) = 1$), we must have $p(x = 3|y = 2) = 2/3$.

In the Monty Hall problem, $p(x)$ is a subjective *prior probability distribution*. It represents *our* (the contestant’s) assessment of the situation *before* the host had opened one of the doors. It is subjective because it may differ from the host’s assessment. (Monty *knows*

¹See Jason Rosenhouse, *The Monty Hall Problem: The Remarkable Story of Math’s Most Contentious Brain Teaser*, 2009, for more than you ever wanted to know about this.

which door is hiding the prize.) It is a prior distribution because these are our beliefs at the beginning of the game. After the host opens door y , these beliefs are modified to the *posterior probability distribution* $p(x|y)$. That is, we *updated* our beliefs to reflect new information. Bayes' Theorem tells us how we should update our beliefs, as summarized in Table 1.

x	Prior	Posterior	
	$p(x)$	$p(x y = 2)$	$p(x y = 3)$
1	1/3	1/3	1/3
2	1/3	0	2/3
3	1/3	2/3	0

Table 1: *Bayesian Updating for the Monty Hall Problem. The second column shows the contestant's prior probability distribution for x (the door hiding the prize) before observing y . The third and fourth columns show the posterior probability distribution for x after observing either $y = 2$ or $y = 3$. The host never opens door 1, which was initially selected by the contestant.*

1.3. Bayesian Statistical Modeling. In this course, we will study statistical modeling. In applied statistics, we have some data and we specify a *probability model* for the data. The model usually depends upon some unknown parameters and the purpose of the analysis is to make inferences about those parameters or to make out of sample predictions. The standard approach has been to use the method of maximum likelihood (or one of its variants) to estimate the parameters. The sampling distribution of the MLE is approximately normal with variance equal to minus the the inverse of the information matrix. This approach works well when there is a lot of data, there aren't too many parameters, and the models aren't too complicated. While the amount of data available seems to be ever increasing, data remains a scarce commodity. The models we wish to entertain grow in complexity, frequently involving latent or missing variables. Modern Bayesian methods are ideally suited for models complex models with many parameters.

In Bayesian statistical modeling, we also start with a likelihood function $p(y|\theta)$, where y denotes the data and θ the unknown parameters. Both y and θ are usually vector or matrix valued, but the dimensions aren't important at this stage. To conduct Bayesian inference, we also need to specify a *prior distribution* $p(\theta)$ for θ . The prior is supposed to represent our subjective beliefs about θ . These may come from earlier studies, from theorizing, or casual observation. We then use Bayes' Theorem (and modern computational methods) to compute the posterior distribution $p(\theta|y)$. Bayesian inference tends to be much simpler than classical inference, with its bag of tricks and special cases. There principles of Bayesian inference are very simple: *Everything is random. Condition on the observables and describe everything else using its posterior probability distribution.*

Bayesian inference, like most things, has some drawbacks. The most obvious is the need for a prior distribution over the parameters. It may be difficult to express our prior beliefs in the form of a probability distribution. The prior distribution is subjective, so, if

your prior differs from mine, you may not agree with my conclusions. However, modern Bayesian inference tries to minimize dependence upon priors. We often use uninformative or “weakly informative” priors, so that the data tend to dominate the priors. We conduct sensitivity analyses to determine whether a wide range of priors can lead to similar conclusions. We will see that in many situations a Bayesian analysis with a non-informative prior distribution yields results similar to classical analyses. However, this is not always the case and Bayesian methods are now applicable to models that were considered difficult or impossible to estimate using MLE not too long ago.

1.4. Notation. Generally, we will not bother to distinguish between probability mass functions and density functions, so either x or y may be discrete, continuous, or some mixture. Integrals should be interpreted appropriately, so, for example, if x were discrete, then $\int p(x, y) dx = \sum_x p(x, y)$. We will also be casual about using the same symbol to denote different density functions: $p(x)$ will denote the density of x and $p(y)$ will denote the density of y , though the two densities could be completely different. If we need to be precise, we will use subscripts, *e.g.* $p_x(x)$ and $p_y(y)$. For example, suppose $y = g(x)$ where g is strictly monotone and differentiable, then

$$p_y(y) = p_x(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right|.$$

This would have been hard to follow without the subscripts. Usually, no confusion will arise and the subscripts can be omitted.

I will try to use Greek letters such as θ for parameters (which are unobservable) and roman letters such as x and y for data (which may be either observed or unobserved). An exception will be finite population sampling, where I will try to adhere to the usual notation (with upper case letters for parameters and lower case letters for sample values). Otherwise, lower case letters will be used for scalars and vectors, upper case letters for matrices. No distinction will be made between random variables and their realized values. In Bayesian statistics, everything is random. The important distinction is between what is observed (and, hence, known) and what is not observed (and, hence, unknown, whether it’s potentially observable data or a parameter).

2. FINITE POPULATION SAMPLING

Sampling is an important tool in the social sciences. Often we need to describe aspects of large populations and it is too costly or time-consuming to study the entire population. Instead, we analyze a sample from that population. The population could be people (*e.g.*, registered voters in an election survey) or things (*e.g.*, news stories in a content analysis).

2.1. Definition and Notation. First, some notation and basic concepts. The *population* or *universe* $U = \{1, \dots, N\}$ is composed of N elements, where N is finite (hence the term “finite population sampling”). A *sample* S is a subset of the population U . For each

element $i \in U$, we can define the *sample inclusion indicator* D_i by

$$D_i = \begin{cases} 1 & i \in S \\ 0 & i \notin S \end{cases}$$

The sample size will be denoted by n , where

$$n = |S| = \sum_{i=1}^N D_i.$$

For each unit i , the *selection probability* is

$$\pi_i = \mathbb{P}(i \in S) = \mathbb{E}(D_i).$$

For a pair of units i and j , the joint sample inclusion probability is

$$\pi_{ij} = \mathbb{P}(i \in S, j \in S) = \mathbb{E}(D_i D_j).$$

Associated with each unit in the population is an attribute y_i (which can be vector-valued), called the *survey variable*. The purpose of sampling is to estimate some feature of the population, such as the *population mean*,

$$\bar{Y} = \frac{1}{N} \sum_{i=1}^N y_i.$$

In most cases, the sample is selected and data on y_i is only collected for sample members, so y_i is unobserved for $i \notin S$. However, the sample mean

$$\bar{y}_S = \frac{1}{n} \sum_{i \in S} y_i = \frac{1}{n} \sum_{i=1}^N D_i y_i$$

can be used as an estimate of $\bar{Y}$. Sometimes, there are *auxiliary variables* x_i which are observed for the entire population from a census or other source. These can be very helpful in estimation and will be discussed later in the course.

2.2. Simple Random Sampling without Replacement. In *simple random sampling without replacement* (SRSWOR), the sample size n is fixed and all samples of n distinct elements are equally likely to be selected. This can be implemented by drawing elements sequentially and removing elements as they are selected until n elements have been chosen. Since there are

$$\binom{N}{n} = \frac{N!}{n!(N-n)!}$$

possible ways to choose n distinct elements from a set of N elements,

$$p(S) = \binom{N}{n}^{-1}.$$

This is called the *sample design*.

Proposition 1. *In simple random sampling without replacement,*

$$\pi_i = n/N \quad \text{for } i = 1, \dots, N.$$

$$\pi_{ij} = \begin{cases} n/N & \text{for } i = j \\ n(n-1)/N(N-1) & \text{for } i \neq j \end{cases}$$

Proof. The inclusion probabilities can be obtained by simple combinatorial arguments, e.g. if $i \in S$, there are $\binom{N-1}{n-1}$ ways to choose the remaining $n-1$ sample elements, so

$$\pi_i = \frac{\binom{N-1}{n-1}}{\binom{N}{n}} = \frac{(N-1)!}{(n-1)!(N-n)!} \cdot \frac{n!(N-n)!}{N!} = \frac{n}{N}.$$

For $i = j$, $\pi_{ij} = \pi_i = n/N$. When $i \neq j$, once i and j have been selected, there are

$$\binom{N-2}{n-2} = \frac{(N-2)!}{(n-2)!(N-n)!}$$

ways to choose the remaining $n-2$ elements and divide by the total number of samples. $\square$

(A slightly slicker proof for the first order inclusion probability: since n is fixed,

$$n = \mathbb{E}(n) = \mathbb{E}\left(\sum_{i=1}^N D_i\right) = \sum_{i=1}^N \pi_i = N\pi_1,$$

using symmetry. Can you find a similar trick to get the second order inclusion probabilities?)

2.3. Sampling Distribution of the Sample Mean. In classical statistics, inference is based upon the *sampling distribution* of estimates. The sampling distribution describes how an estimate, such as the sample mean $\bar{y}_S$, varies in repeated sampling from the same population. In practice, we usually have just a single sample (not repeated samples), but the properties of the sampling distribution can be inferred analytically.

In finite population sampling, the population is fixed, so $y_1, \dots, y_N$ are fixed (but unknown, at least for the unsampled units). Randomness in $\bar{y}_S$ is due to S , not the survey variable. The expected value of $\bar{y}_S$ is obtained by computing the average of $\bar{y}_S$ over the $\binom{N}{n}$ possible samples $S \subset U$.

Proposition 2. *Under SRSWOR, $\mathbb{E}(\bar{y}_S) = \bar{Y}$.*

Proof.

$$\mathbb{E}(\bar{y}_S) = \mathbb{E}\left(\frac{1}{n} \sum_{i=1}^N D_i y_i\right) = \frac{1}{n} \sum_{i=1}^N \mathbb{E}(D_i) y_i = \frac{1}{n} \sum_{i=1}^N \frac{n}{N} y_i = \frac{1}{N} \sum_{i=1}^N y_i = \bar{Y}.$$

$\square$

The expression for the variance requires more algebra and may look a little different from what you expected.

Proposition 3. *Under SRSWOR,*

$$\mathbb{V}(\bar{y}_S) = (1 - f) \frac{\sigma^2}{n},$$

where $f = n/N$ is the sampling fraction and $\sigma^2 = (N - 1)^{-1} \sum_{i=1}^N (y_i - \bar{Y})^2$. An unbiased estimator of $\mathbb{V}(\bar{y}_S)$ is

$$\hat{\mathbb{V}}(\bar{y}_S) = (1 - f) \frac{\hat{\sigma}^2}{n},$$

where $\hat{\sigma}^2 = (n - 1)^{-1} \sum_{i \in S} (y_i - \bar{y}_S)^2$.

Proof. Since $\mathbb{V}(\bar{y}_S) = \mathbb{E}(\bar{y}_S^2) - [\mathbb{E}(\bar{y}_S)]^2 = \mathbb{E}(\bar{y}_S^2) - \bar{Y}^2$ and

$$\begin{aligned} \mathbb{E}(\bar{y}_S^2) &= \mathbb{E} \left(\frac{1}{n^2} \sum_{i=1}^N \sum_{j=1}^N D_i D_j y_i y_j \right) \\ &= \frac{1}{n^2} \left(\sum_{i=1}^N \pi_i y_i^2 + \sum_{i \neq j} \pi_{ij} y_i y_j \right) \\ &= \frac{1}{n^2} \left(\frac{n}{N} \sum_{i=1}^N y_i^2 + \frac{n(n-1)}{N(N-1)} \sum_{i \neq j} y_i y_j \right) \\ &= \frac{1}{nN} \left[\left(1 - \frac{n-1}{N-1} \right) \sum_{i=1}^N y_i^2 + \frac{n-1}{N-1} \left(\sum_{i=1}^N y_i \right)^2 \right] \\ &= \frac{1}{nN(N-1)} \left[(N-n) \sum_{i=1}^N y_i^2 + (n-1)N^2 \bar{Y}^2 \right] \end{aligned}$$

we obtain

$$\mathbb{V}(\bar{y}_S) = \frac{N-n}{nN(N-1)} \left(\sum_{i=1}^N y_i^2 - N\bar{Y}^2 \right) = (1-f) \frac{\sigma^2}{n}.$$

Unbiasedness of $\hat{\mathbb{V}}(\bar{y}_S)$ is left as an exercise. □

The sampling fraction f appears in the variance formula because we are sampling without replacement. When $n = N$, the sample mean is the population mean. In most applications, the sampling fraction is small and can be ignored, so the variance becomes the customary σ^2/n , familiar from assuming i.i.d. observations. When sampling without replacement, the observations are slightly negatively correlated, which reduces sampling variability a little.

2.4. Confidence Intervals. Point estimates, such as the sample mean, are subject to error and, for many purposes, it is more useful to have an interval which provides a range of values likely to contain the parameter being estimated. For example, the interval $[\bar{y}_S \pm 1.96 \text{ s.e.}(\bar{y}_S)]$ is often used to provide an approximate 95 percent confidence interval for the population mean $\bar{Y}$. The conditions under which the normal approximation is reasonable are difficult to state.² The usual complications in describing the interval arise. For example, is the confidence level the probability that the interval contains the population mean? The limits of the interval are random and have a long run relative frequency interpretation, but strict frequentists would reject any probability statement about a single interval as being meaningless.

In general, for a two-sided $100(1 - \alpha)\%$ confidence interval, we use the $1 - \alpha/2$ quantile of the sampling distribution as the multiplier. For small samples, the use of a quantile of the Student t distribution with $n - 1$ degrees of freedom is often recommended in place of the normal quantile. The justification for using the t distribution is usually fuzzy. For example, Cochran, *Sampling Techniques*, 3rd ed., 1977, p. 27, states that “The t distribution holds exactly only if the observations y_i are themselves normally distributed and N is infinite. Moderate departures from normality do not affect it greatly. For small samples with very skew distributions, special methods are needed.” Thus, the confidence limits will be $[\bar{y}_S \pm t_{1-\alpha/2, n-1} \text{ s.e.}(\bar{y}_S)]$ for a $100(1 - \alpha)\%$ confidence interval where $t_{p, \nu}$ is the p th quantile of the Student t distribution with ν degrees of freedom.

3. EXAMPLE: SAMPLING A DICHOTOMOUS ATTRIBUTE

We are long overdue an example.

Example 1. *We draw a SRSWOR of size $n = 5$ from a population of $N = 20$ persons and ask each selected person who they intend to vote for. We obtain a proportion $\bar{y}_S = 4/5$ who say they are Democrats.*

In this population, the survey variable is dichotomous: $y_i = 0$ or 1 for all i . The sample mean is the proportion of sample observations for which $y_i = 1$ and the population mean is the corresponding proportion of the population having $y_i = 1$.

3.1. Classical Frequentist Analysis. In the standard analysis, we would report the point estimate $\bar{y}_S = 0.8$ along with a 95% confidence interval for $\bar{Y}$.

²The usual approach is to embed the sample design in a sequence of designs with $n/N \rightarrow f$ as $n, N \rightarrow \infty$. Some additional assumptions about $y_1, \dots, y_N$. See Hajek, *Sampling from a Finite Population*, 1981, or Fuller, *Sampling Statistics*, 2009, for details. Section 2.15 of Cochran, *Sampling Techniques*, 3rd ed., 1977, contains some useful advice.

To compute the confidence interval, note that $f = n/N = 5/20 = 1/4$ and the sample variance is

$$\begin{aligned}\hat{\sigma}^2 &= \frac{1}{n-1} \sum_{i \in S} (y_i - \bar{y})^2 = \frac{n\bar{y}_S(1 - \bar{y}_S)^2 + n(1 - \bar{y}_S)(0 - \bar{y}_S)^2}{n-1} \\ &= \frac{n\bar{y}_S(1 - \bar{y}_S)}{n-1} = \frac{5}{4} \times \frac{4}{5} \times \frac{1}{5} = \frac{1}{5}.\end{aligned}$$

Then the standard error of $\bar{y}_S$ is

$$\text{s.e.}(\bar{y}_S) = \sqrt{\frac{(1-f)\hat{\sigma}^2}{n}} = \sqrt{\frac{(1-1/4) \times (1/5)}{5}} = \sqrt{0.03} \doteq 0.17$$

We can form a 95 percent confidence interval for $\bar{Y}$:

$$[0.8 \pm 2.78 \times 0.17] = [0.8 \pm 0.48] = [0.32, 1.28]$$

The multiplier of 2.78 is the 0.975 quantile of the Student t distribution with $n-1 = 4$ degrees of freedom.

Here is how to compute the analysis using R:

```
> N <- 20
> y <- c(1,1,1,1,0)
> n <- length(y)
> ybar <- mean(y)
> se.ybar <- sqrt((1 - n/N) * var(y) / n)
> ybar; se.ybar; qt(.975, df=4)
[1] 0.8
[1] 0.1732
[1] 2.776
> ybar + qt(c(.025, .975), df=n-1) * se.ybar
[1] 0.3191 1.2809
```

We cannot use the `t.test` function, since it doesn't make the finite population correction to the standard error.

The same analysis can be performed using the R `survey` package, though it is overkill for this problem.

```
> library(survey)
> srswor <- svydesign(ids=~1, fpc=rep(n/N, n), data=data.frame(y=y))
> svymean(~y, design=srswor)
      mean      SE
y  0.8 0.1732
```

The `survey` library requires the user to specify a `svydesign` object. The argument `ids=~1` says that there are no clusters, `fpc` gives the finite population correction (it needs to have as many rows as the data). The `data` argument takes a `data.frame` object. The function `svymean` computes the sample mean and its standard error. The first argument to `svymean` is a formula specifying which variables to calculate the mean for. See Lumley, *Complex Surveys: A Guide to Analysis Using R*, 2010, and the package documentation on CRAN

for details. (Incidentally, the generic `confint` function uses normal rather than Student t quantiles, so it gives slightly narrower intervals.)

For $n = 5$ and $N = 20$, the actual coverage of these intervals is not particularly close to 95% for most values of $\bar{Y}$, as shown in Figure 1 below. The computation uses the hypergeometric distribution described in the next section.

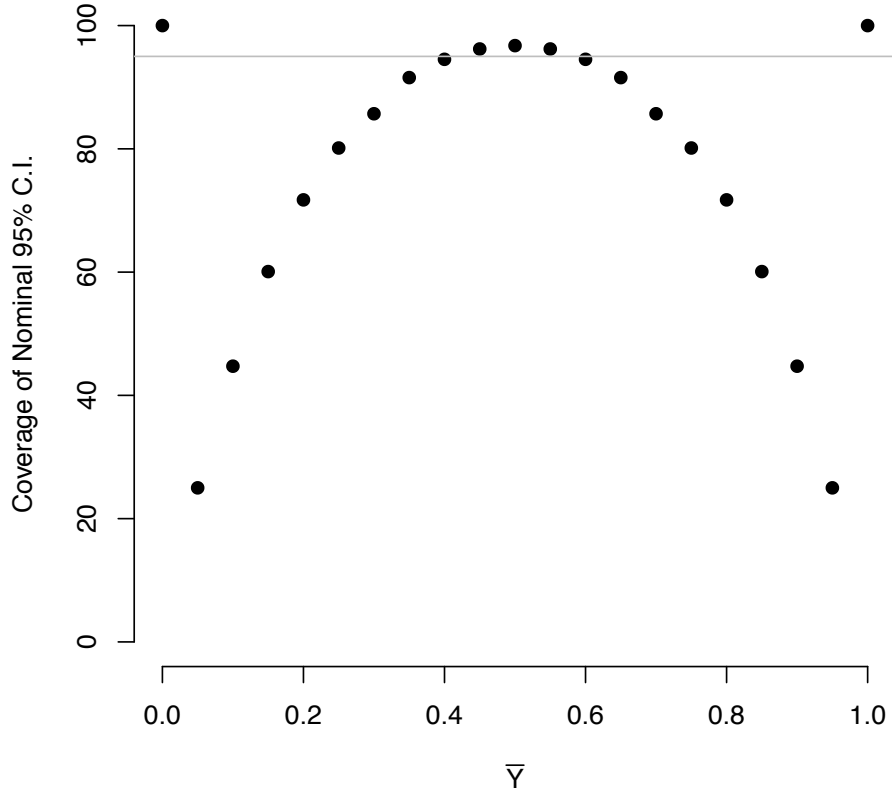


FIGURE 1. *The dots indicate the actual coverage rates of nominal 95% confidence intervals for $\bar{Y} = 0, 1/20, \dots, 19/20, 1$. The horizontal gray line is at 95%. If the nominal coverage was correct, all points should be on (or near) this line. In fact, except for $0.4 \leq \bar{Y} \leq 0.6$, the nominal coverage level is quite misleading.*

3.2. Bayesian Analysis. There are some confusing aspects to the classical frequentist analysis. For example, what are the parameters? what is the likelihood function? what is the MLE? These are also issues for the Bayesian analysis of the finite population sampling problem, but we can sidestep these problems when the survey variable is dichotomous. We will let $\bar{Y}$ (the population proportion) be the parameter and derive the likelihood for $\bar{Y}$

when the sample proportion is $\bar{y}$. (We drop the S subscript on $\bar{y}$ since it doesn't make any difference which sample $\bar{y}$ came from.)

Let $M = N\bar{Y}$ denote the number of Democrats in the population of $N = 20$ units. Let $m = n\bar{y}_S$ denote the number of Democrats in the sample of $n = 5$ units. It is easier to work with m and M instead of $\bar{y}$ and $\bar{Y}$, but since n and N are known and fixed, they carry the same information. Note that $N - M$ is the number of Republicans in the population and $n - m$ is the number of Republicans in the sample. Since there can't be more Democrats in the sample than in the population and the same holds for Republicans, we must have $M \geq m$ and $N - M \geq n - m$. On the other hand, both m and $n - m$ must be non-negative, and neither can exceed n . Combining these restrictions gives

$$\max\{0, n - N + M\} \leq m \leq \min\{n, M\}$$

The distribution of m given M is *hypergeometric*:

$$p(m|M) = \frac{\binom{M}{m} \binom{N-M}{n-m}}{\binom{N}{n}}.$$

The likelihood in terms of $\bar{Y}$ and $\bar{y}_S$ is obtained by substituting $N\bar{Y}$ for M and $n\bar{y}$ for m in this formula gives

$$p(\bar{y}|\bar{Y}) = \frac{\binom{N\bar{Y}}{n\bar{y}} \binom{N(1-\bar{Y})}{n(1-\bar{y})}}{\binom{N}{n}}.$$

Finally, we need a prior for $\bar{Y}$. I'm fairly certain that $30\% \leq \bar{Y} \leq 100\%$ and a $\text{Normal}(0.7, 0.15^2)$ distribution puts 95% of its probability on this interval. However, the prior distribution for $\bar{Y}$ needs to be discrete, since the possible values are the $N + 1$ values $\{0, 1/N, \dots, (N-1)/N, 1\}$, so a reasonable choice would be

$$p(\bar{Y} = k/N) = \frac{\phi_k}{\sum_{j=0}^N \phi_j} \quad \phi_k = \phi\left(\frac{k - 0.7}{0.15}\right)$$

for $k = 0, 1, \dots, N$, and $\phi(z)$ is the standard normal density function. This makes the shape of the distribution normal, concentrates most of the probability between 0.4 and 0.7, and ensures that the probabilities sum to one. What more could you ask for?

The posterior distribution of $\bar{Y}$ given $\bar{y}$ is

$$\begin{aligned} p(\bar{Y} = k/N|\bar{y}) &= \frac{p(\bar{Y}) p(\bar{y}|\bar{Y} = k/N)}{\sum_{j=0}^N p(\bar{Y} = j/N) p(\bar{y}|\bar{Y} = j/N)} \\ &= \frac{\phi_k \binom{k}{n\bar{y}} \binom{N-k}{n(1-\bar{y})}}{\sum_{j=0}^N \phi_j \binom{j}{n\bar{y}} \binom{N-j}{n(1-\bar{y})}} \end{aligned}$$

The prior, likelihood, and posterior are shown in Figure 2.

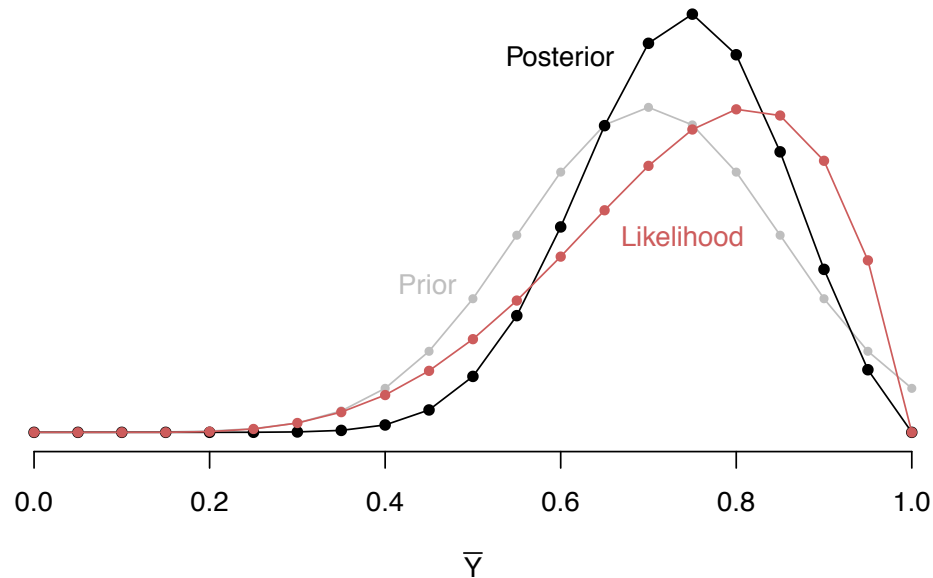


FIGURE 2. The posterior distribution of $\bar{Y}$ is a compromise between the prior distribution and the likelihood with a posterior mean of 0.73. The MLE of $\bar{Y}$ is 0.8.

We can form a 95% *credible interval* for $\bar{Y}$ (the Bayesian analog of a 95% confidence interval) by inspection of the cumulative posterior probabilities, except that we really do mean that $\mathbb{P}(0.50 \leq \bar{Y} \leq 0.90 | \bar{y}) = 0.96$.

Here is some R code to perform the computations and graph the prior, likelihood, and posterior. (The likelihood needs to be normalized to sum to one and fit on the graph.)

```
> Ybar <- (0:N) / N
> prior <- dnorm(Ybar, mean=0.7, sd=0.15)
> prior <- prior / sum(prior)
> likelihood <- dhyper(sum(y), N*Ybar, N*(1-Ybar), n)
> posterior <- prior * likelihood
> posterior <- posterior / sum(posterior)
> sum(Ybar * posterior)      # posterior mean
[1] 0.7315
> data.frame(Ybar=Ybar, cumprob=cumsum(posterior))
  Ybar cumprob
1  0.00 0.000e+00
```

```

2  0.05  0.000e+00
3  0.10  0.000e+00
4  0.15  0.000e+00
5  0.20  1.664e-06
6  0.25  2.408e-05
7  0.30  1.855e-04
8  0.35  9.900e-04
9  0.40  4.048e-03
10 0.45  1.335e-02
11 0.50  3.657e-02
12 0.55  8.503e-02
13 0.60  1.703e-01
14 0.65  2.977e-01
15 0.70  4.593e-01
16 0.75  6.330e-01
17 0.80  7.898e-01
18 0.85  9.063e-01
19 0.90  9.740e-01
20 0.95  1.000e+00
21 1.00  1.000e+00
> sum(posterior[0.50 <= Ybar & Ybar <= 0.90])
[1] 0.9607
> likelihood <- likelihood / sum(likelihood)
> plot(x=Ybar, y=prior, type="p", pch=19, cex=0.6,
+      xlim=c(0,1), xlab=expression(bar(Y)),
+      ylim=c(0,.2), ylab="", yaxt="n", bty="n",
+      col="grey")
> lines(x=Ybar, y=prior, col="grey", cex=0.7)
> points(x=Ybar, y=posterior, col="black", cex=0.8, pch=19)
> text(x=0.5, y=0.06, label="Prior", col="grey", pos=2)
> lines(x=Ybar, y=posterior, col="black")
> text(x=0.68, y=0.155, label="Posterior", col="black", pos=2)
> points(x=Ybar, y=likelihood, col="indianred", pch=19, cex=0.7)
> lines(x=Ybar, y=likelihood, col="indianred")
> text(x=0.65, y=0.08, label="Likelihood", col="indianred", pos=4)

```